

Nonconvex penalties for graphical continuous Lyapunov models

Simulation campaign, October 2026: verdicts and evidence

Joon Kim

9 October 2026

Model and estimators

GCLM $M\Sigma + \Sigma M^\top + C = 0$, M stable; graph: edge $j \rightarrow i \iff M_{ij} \neq 0$ ($i \neq j$)

Every estimator: $\hat{M}_\lambda \in \arg \min_M L(M) + \sum_{i \neq j} P_\lambda(|M_{ij}|)$, for 100 values of λ ; diagonal of M unpenalised

- **direct loss** [1], all main results: $L(M) = \frac{1}{2} \|M\hat{\Sigma} + \hat{\Sigma}M^\top + C\|_F^2$
- **log-likelihood loss** [2], only at $p = 10$: $L(M) = \log \det \Sigma(M) + \text{tr}(\Sigma(M)^{-1}\hat{\Sigma})$, $\Sigma(M)$ from the GCLM equation
- **lasso**: $P_\lambda(t) = \lambda t$
- **MCP** [5], $\gamma = 3$: $P_\lambda(t) = \begin{cases} \lambda t - t^2/(2\gamma), & t \leq \gamma\lambda \\ \gamma\lambda^2/2, & t > \gamma\lambda \end{cases}$
- **SCAD** [4], $\gamma = 3.7$: the same idea
- **adaptive lasso** [6], convex: $P_\lambda(|M_{ij}|) = \lambda w_{ij} |M_{ij}|$
 $w_{ij} = \max_{kl} |\tilde{M}_{kl}| / |\tilde{M}_{ij}|$, $\tilde{M}$: lasso solution at $\lambda_{\min}$

sparse $\rightarrow$ dense: $\lambda \downarrow$



Computing a path: the 100 problems are solved one after another in the direction of the arrow.
Each solve (FISTA [9]) is initialised with the solution of the previous one.

Data, and from a path to one graph

True models [1, §5]: $M_{ij} = \omega_{ij}\varepsilon_{ij}$, $\omega_{ij} \sim \text{Bern}(k/p)$, $\varepsilon_{ij} \sim N(0, 1)$; $M_{ii} = -\sum_{j \neq i} |M_{ij}| - |\varepsilon_{ii}|$
 $p = 10, 20$ (and 15 to 50), $k = 1, \dots, 4$, four true C ; $n = 10^3, 10^4$ or ∞ ; standardised; 400 graphs per p
 C in the fit: $2I$ (Detting's pipeline) or $2 \text{diag}(s_1^{-2}, \dots, s_p^{-2})$ (rescaled: correct after standardising)

A path gives 100 graphs, one per λ .

Two ways to choose one of them:

- **oracle:** $\hat{\lambda} = \arg \max_{\lambda} F_1(\hat{M}_{\lambda})$ uses the truth
- **score:** the support S on the path with the lowest
$$\text{score}(S) = \underbrace{n[\log \det \Sigma_S + \text{tr}(\Sigma_S^{-1} \hat{\Sigma})]}_{\text{loss}} + \underbrace{\text{pen}(S)}_{\text{penalty}}$$

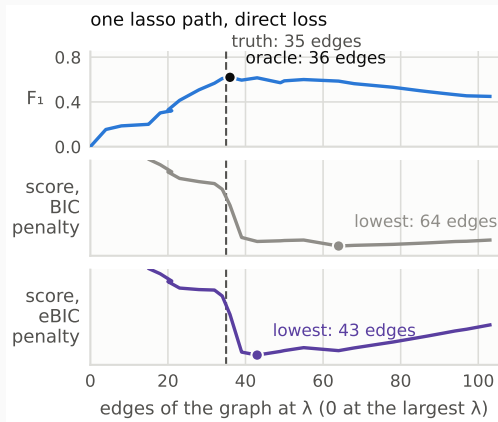
M_S : least-squares refit of the direct loss on S

Σ_S : solves $M_S \Sigma_S + \Sigma_S M_S^T + C = 0$ $n = \infty$: $n = 10^6$

- **BIC:** $\text{pen}(S) = (p + |S|) \log n$
- **eBIC** [1]:

$$\text{pen}(S) = (p + |S|) \log n + 4\gamma |S| \log p, \quad \gamma = 1$$

Measured: directed F_1 of the chosen graph; AUPR of the path



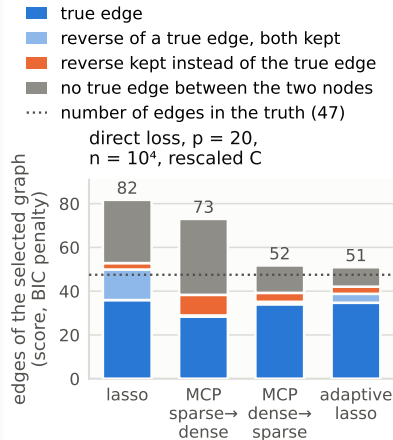
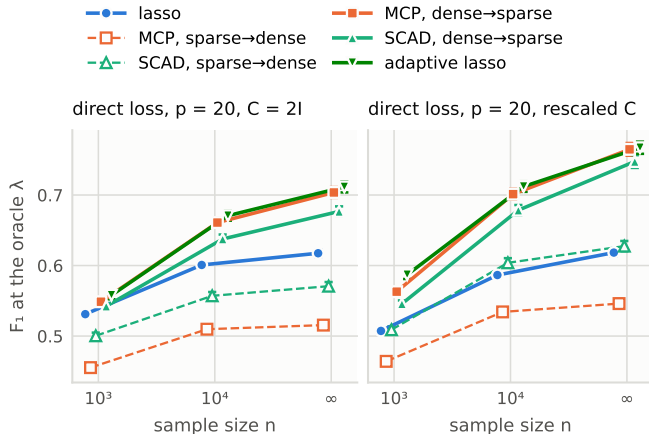
Direct loss

- V1** With $C = 2I$, sparse $\rightarrow$ dense MCP and SCAD lose to the lasso.
They lock in wrong edge directions early on the path.
- V2** Dense $\rightarrow$ sparse MCP and SCAD beat the lasso, or tie at small p .
- V3** The convex adaptive lasso, from the same dense start, does as well.
So the dense start matters, not the nonconvexity.
- V4** The advantage of the dense start grows with p .
- V5** At large p the score with the BIC penalty selects too many edges.
The eBIC penalty is better there; at small p it drops true edges and is worse.
- V6** A greedy search from the selected graph closes the lasso's gap to dense $\rightarrow$ sparse.
Started from random graphs instead, the search needs many starts to get as far.

Log-likelihood loss, only at $p = 10$ (backup slide)

Sparse $\rightarrow$ dense MCP loses to the lasso; dense $\rightarrow$ sparse MCP does not beat it for finite n .

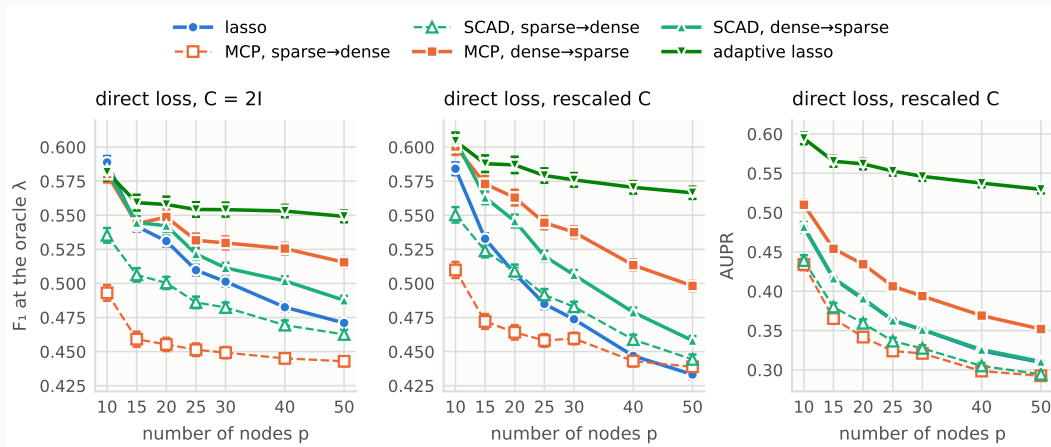
V1-V3, direct loss: the two path orders and the adaptive lasso



Left: dashed (sparse $\rightarrow$ dense) lies below the lasso; solid (dense $\rightarrow$ sparse) lies above.

Right: sparse $\rightarrow$ dense MCP often keeps the reverse of a true edge instead of the edge (orange).

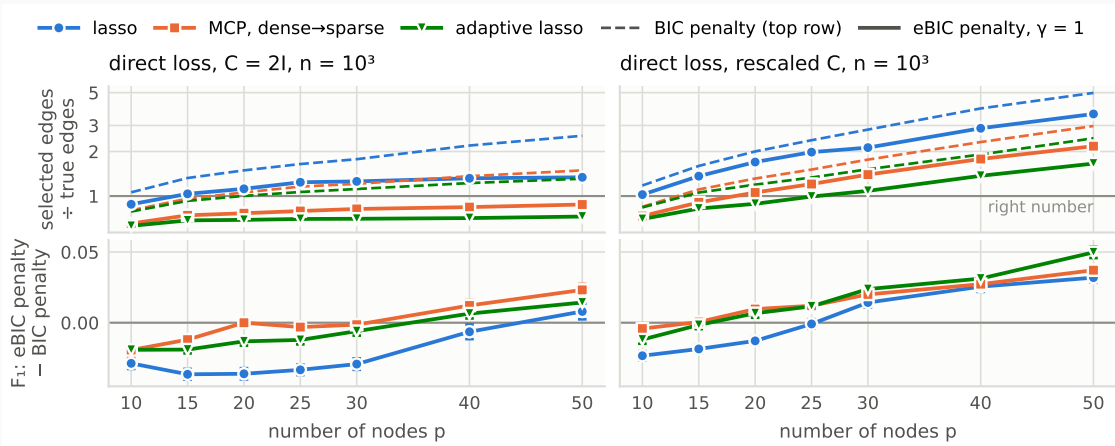
V3-V4, direct loss: the advantage grows with p ($n = 10^3$)



The lasso degrades with p ; the adaptive lasso barely does.

At $p = 40, 50$ there are more parameters than observations, and the gains are largest there.

V5, direct loss: the BIC penalty against Dettling's eBIC penalty ($n = 10^3$)



Top: the BIC penalty (dashed) gives too many edges at large p ; the eBIC penalty (solid) fewer.

Bottom: the extra $4\gamma \log p$ per edge raises F_1 only where the BIC penalty gives too many edges.

V6, direct loss: the greedy search

Score of a graph G : fit M on G by least squares on the direct loss ([3]: maximise the likelihood), then $\text{score}(G)$ as on slide 2, with the BIC penalty.

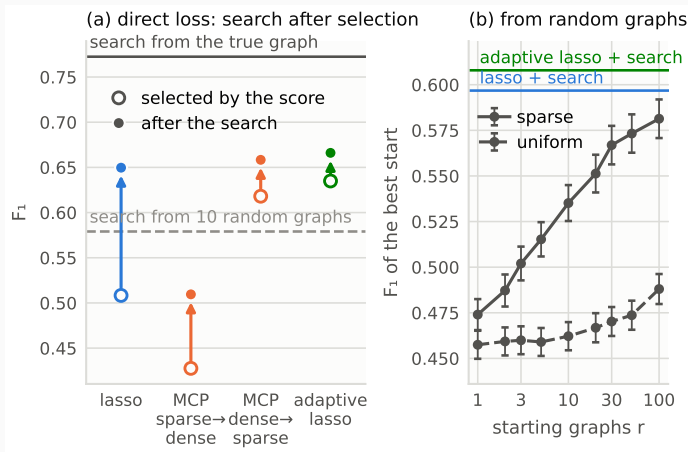
Search from a graph S , as in [3]:

1. score every graph that differs from S by one edge (added, deleted, reversed)
2. if the best of them has a lower score than S , it becomes S ; repeat

Where the search starts

- (a) the graph the score selects on a path
- (b) r random graphs and the empty graph, one search each; the end graph with the lowest score is kept ([3]: $r = 300$)

Random graphs: sparse, edge probability from $U[0, 0.3]$; uniform, $1/2$ per edge [8].

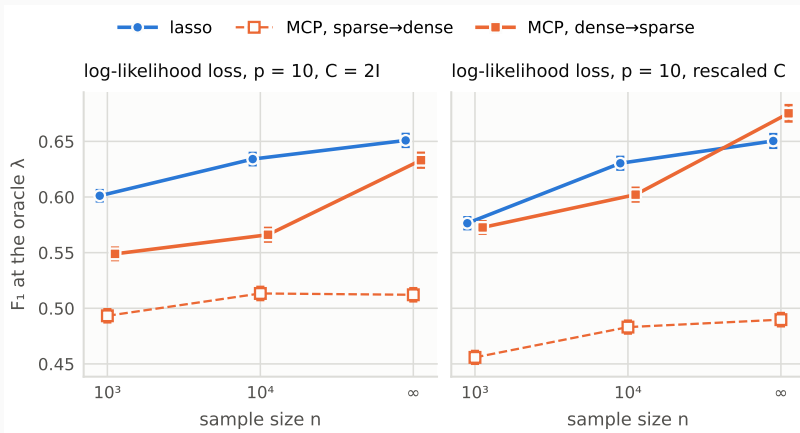


(a) $p = 20$, (b) $p = 10$; both $n = 10^4$, rescaled C , direct loss.

- **Running:** the eBIC penalty inside the search for every setting; 300 random starting graphs
- **Next:** the log-likelihood loss at $p = 20$ for both C ; the likelihood refit in the search
- **Open:** a theory for why sparse $\rightarrow$ dense paths lock in wrong edge directions

- [1] Dettling, Drton & Kolar (2024). On the lasso for graphical continuous Lyapunov models. *CLearR 2024*.
- [2] Varando & Hansen (2020). Graphical continuous Lyapunov models. *UAI 2020*.
- [3] Améndola, Dettling, Drton, Onori & Wu (2020). Structure learning for cyclic linear causal models. *UAI 2020*.
- [4] Fan & Li (2001). Variable selection via nonconcave penalized likelihood and its oracle properties. *JASA* 96.
- [5] Zhang (2010). Nearly unbiased variable selection under minimax concave penalty. *Ann. Statist.* 38.
- [6] Zou (2006). The adaptive lasso and its oracle properties. *JASA* 101.
- [7] Chen & Chen (2008). Extended Bayesian information criteria for model selection with large model spaces. *Biometrika* 95.
- [8] Nowzohour, Maathuis, Evans & Bühlmann (2017). Distributional equivalence and structure learning for bow-free acyclic path diagrams. *Electron. J. Stat.* 11.
- [9] Beck & Teboulle (2009). A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM J. Imaging Sci.* 2.

Backup: the log-likelihood loss ($p = 10$)



Dashed (sparse $\rightarrow$ dense) lies below the lasso; solid (dense $\rightarrow$ sparse) is above it only at $n = \infty$, rescaled C .
Dense $\rightarrow$ sparse starts here from the exact fit $M = -\frac{1}{2}C\hat{\Sigma}^{-1}$ [2], not from the lasso solution.